Filbin et al. 2018 — Pediatric H3-K27M Mutant Glioma

SmartSeq2 (TPM) · 6 patients · 2,259 malignant cells · Brain

Geometry: BRANCHING

OPC-like progenitor hub → AC-like + OC-like glial arms

GATE 1 GO

Harmony(θ=10) integrates 6 patients (iLISI 3.77/6) on one connected manifold; no islands → mixed AND continuous.

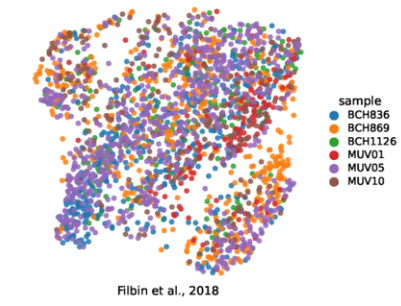
GATE 2 GO

OPC-hub → AC-like (Gast-leaning) + OC-like arms; Crest terminal, no inversion onto the Gast arm.

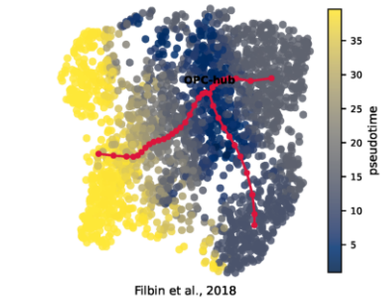
GATE 3 NO-GO

Metric validated on Puram (r=−0.47). Non-cycling hub shows NO Gast↔pSom anti-corr (fork-cluster +0.27; OPC-id −0.03; earliest pre-commitment bin +0.03). Branching is glial, not embryonic.

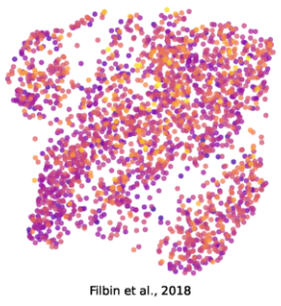
Samples (integration)



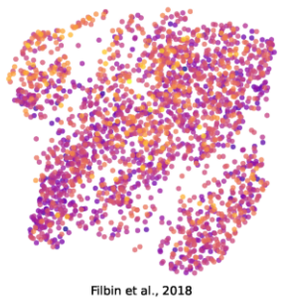
PPT tree + pseudotime



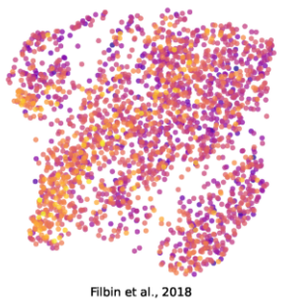
Gastrulation



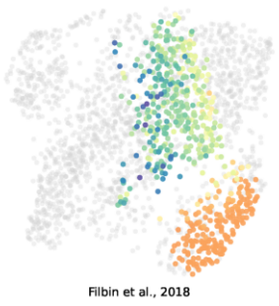
Somitogenesis



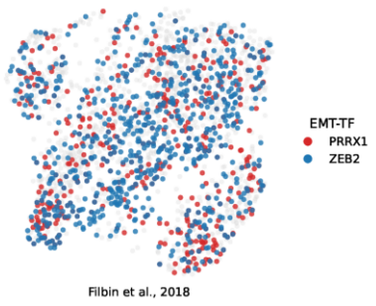
Neural Crest



Module corr (Gate 3: hub≈0)



EMT-TFs: PRRX1 / ZEB2




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PARAMETERS (derived)
• HVGs: 3,779 overdispersed; PCA 50
• Harmony θ=10 on 'sample'; n_pcs=20
• UMAP spread=1.3; islands removed=0
• PPT TREE on X_pca_harmony:
   ndims=10, Nodes=40, λ=3000, σ=0.5
   → 3 tips / 1 fork (stable Y)
• Root: OPC-progenitor fork (node 27)
   developmental origin (not Crest/cycling)

HUB (by role)
• Non-cycling OPC convergence zone
• Gate-3 r ≈ 0 (no cross-repression)
   whole hub +0.12 → non-cyc ≈ 0

MODULE MEANS (raw)
• Gast +0.018  pSom +0.226  Crest +0.100
• Gast-arm = AC ; Crest not on Gast arm

EMT-TFs: PRRX1 (AC) · ZEB2 (OC)
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VERDICT — Branching glial hierarchy (OPC → AC/OC): Gate 1 GO, Gate 2 GO, Gate 3 NO-GO. The manifold genuinely bifurcates, but the embryonic Gast↔pSom cross-repression axis does NOT reproduce at the non-cycling convergence hub — the bifurcation is a glial developmental program (Filbin 2018), not the Puram embryonic competition. Chain stops at 04_correlation; EMT-TF step = 05_expression.